

Riya Gupta

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LinkedIn | GitHub | LeetCode

EDUCATION

Thapar Institute of Engineering and Technology

B.E. Computer Engineering — CGPA: 9.16

- Merit Scholarship Holder

Patiala, India

August 2023 – Present

SKILLS

Languages:	JavaScript (ES6+), Python, SQL, C/C++, Unix Scripting, HTML5, CSS3
Frameworks:	React.js, Node.js, Express, LangChain, LangGraph
Infrastructure:	Docker, AWS, Terraform (IaC), GitHub Actions CI/CD
Databases:	MongoDB, SQL (MySQL)
Coursework:	Data Structures & Algorithms, OOP, Computer Networks, OS, Cloud Computing, Machine Learning

PROJECTS

Autonomous AI Integration Agent

GitHub

LangGraph, LangChain, Groq LLM API, Node.js

Designed an autonomous LLM-powered agent to perform multi-step analysis workflows using ReAct-based reasoning loops.

- **Built an end-to-end LLM API integration pipeline** using **LangGraph and ReAct-based reasoning**, orchestrating multi-step automated workflows and enabling **intelligent task execution** across a production backend system.
- **Engineered an API gateway layer** with **request throttling, rate limiting, and retry mechanisms** to manage LLM inference traffic, achieving **98% pipeline reliability** under strict API quota constraints.
- **Integrated Groq inference for low-latency AI processing**, reducing task completion latency by **10×** while implementing **API failover and distributed request handling** to maintain system stability at scale.

ScaleOps: Cloud Infrastructure & API Platform

GitHub

Node.js, Terraform, GitHub Actions, AWS, Docker

Designed a production-grade cloud architecture to automate infrastructure provisioning and manage containerized deployments.

- **Architected and deployed cloud infrastructure on AWS using Terraform (IaC)**, implementing **custom VPCs, subnets, and security groups** to enforce network isolation and security standards across reproducible environments.
- **Designed a GitHub Actions CI/CD pipeline** integrating automated Jest unit testing and Docker image builds, **eliminating manual deployment overhead** and enforcing code quality gates before release.
- **Orchestrated containerized Node.js deployments on AWS EC2** using Docker and PM2, ensuring **high availability** through **automated crash recovery** and self-healing runtime patterns.

ELC Project Application System

GitHub

React.js, Node.js, Express, MongoDB, Firebase Auth

Built a platform for student-faculty project collaboration, delivering role-based access control and real-time communication.

- **Designed and implemented RESTful APIs** in Node.js and Express for project discovery, application tracking, and status management, enabling **structured data exchange across user roles**.
- **Integrated Firebase Authentication with RBAC** to enforce role-specific data access boundaries for students and faculty, ensuring **secure, isolated workflows** across all system interactions.
- **Built real-time collaboration features** including event-driven notifications and asynchronous in-app messaging using **Socket.io**, improving cross-team coordination and response turnaround.

EXPERIENCE

Thapar Literary Society

Patiala, India

Joint Secretary (Formerly: Convener, Joint Convener)

November 2023 – May 2026

- Led a **cross-functional team of 15+ members** to plan and execute large-scale university events, managing end-to-end logistics, timelines, and stakeholder communication across departments.
- **Documented and standardized editorial workflows** as Editor and Designer for *Avant Garde 2025*, the society's annual magazine, ensuring consistent delivery quality across contributors.